Chapter 1: Research, Terminology and theories

Lectures 1-2

What is research?

Research is a systematic process of gathering, analyzing, and interpreting information to establish facts and reach new conclusions.

It involves a structured approach to investigating a topic, whether it's a problem, a phenomenon, or a theory, with the goal of expanding knowledge and understanding.

Why is research important in Social Sciences?

Research allows the generation of evidence based information to help us understand social problems, influence policy and find solutions to issues

Research helps us advance knowledge which can be used to address problems

Research assists in the understanding of social realities empirically

Research allows decision making based on concrete evidence of social behaviour

Research enhances societal development by providing an indepth understanding of human behaviour

Terms in research

Concept

Definition: An abstract idea or general notion that represents a phenomenon. Example: "Justice", "Gender Equality", "Poverty", or "Freedom".

Variable

Definition: A measurable representation of a concept that can vary or change.

Types: Independent Variable: The one manipulated or categorized to observe its effect.

Dependent Variable: The one affected or measured.

Control Variable: Held constant to ensure validity.

Example: In a study on education and income, *education level* is an independent variable, and *income* is the dependent variable.

Fact

Definition: A statement that can be objectively verified through observation or evidence. Example: "Water boils at 100°C at sea level."

Theory

Definition: A well-substantiated explanation of some aspect of the natural or social world that is based on a body of evidence and has stood up to repeated testing. Example: Theory of Evolution, Feminist Theory, Social Learning Theory.

Proposition

Definition: A statement that expresses a judgment or opinion about the relationship between concepts or variables. Example: "Increased parental involvement improves student academic performance."

Hypothesis

Definition: A specific, testable prediction about the expected relationship between variables.

Types:

- Null Hypothesis (H_0): Assumes no relationship or effect.
- Alternative Hypothesis (H_1): Assumes a relationship or effect.

Example: "Higher stress levels will lead to decreased academic performance."

Characteristics of theories

Empirical Foundation

- Based on observation and evidence.
- Social science theories are grounded in real-world data collected through qualitative or quantitative research.

Systematic and Organized

- Theories provide a structured framework for understanding social phenomena.
- They organize concepts and relationships in a logical way.

Explanatory and Predictive Power

- They aim to explain why certain social events or behaviors occur.
- Some theories also have predictive capacity, suggesting what might happen under certain conditions.

Characteristics of theories

Testability

- Good theories can be tested and evaluated using research methods.
- They generate hypotheses that can be confirmed or refuted through empirical studies.

Abstract and Generalizable

- Theories are often abstract representations of reality.
- They strive for generalizability across different contexts, populations, or times (though not always universally applicable).

Provisional and Evolving

- Theories in social science are not absolute.
- They can be revised, refined, or replaced as new evidence and perspectives emerge.

Characteristics of theories

Value-Informed

- Unlike natural sciences, social sciences may reflect social values, ideologies, or ethical concerns.
- Theories may be influenced by the theorist's perspective (e.g., Marxist, feminist, postcolonial).

Contextual Sensitivity

- They consider the historical, cultural, political, and economic contexts in which social actions and institutions are embedded.

Multidimensional

- Social science theories often involve multiple variables and complex interactions (e.g., race, class, gender).
- They address intersections and overlapping systems of influence.

Practical Relevance

- Many social science theories aim to inform policy, practice, or social change.
- They help diagnose social problems and propose solutions.